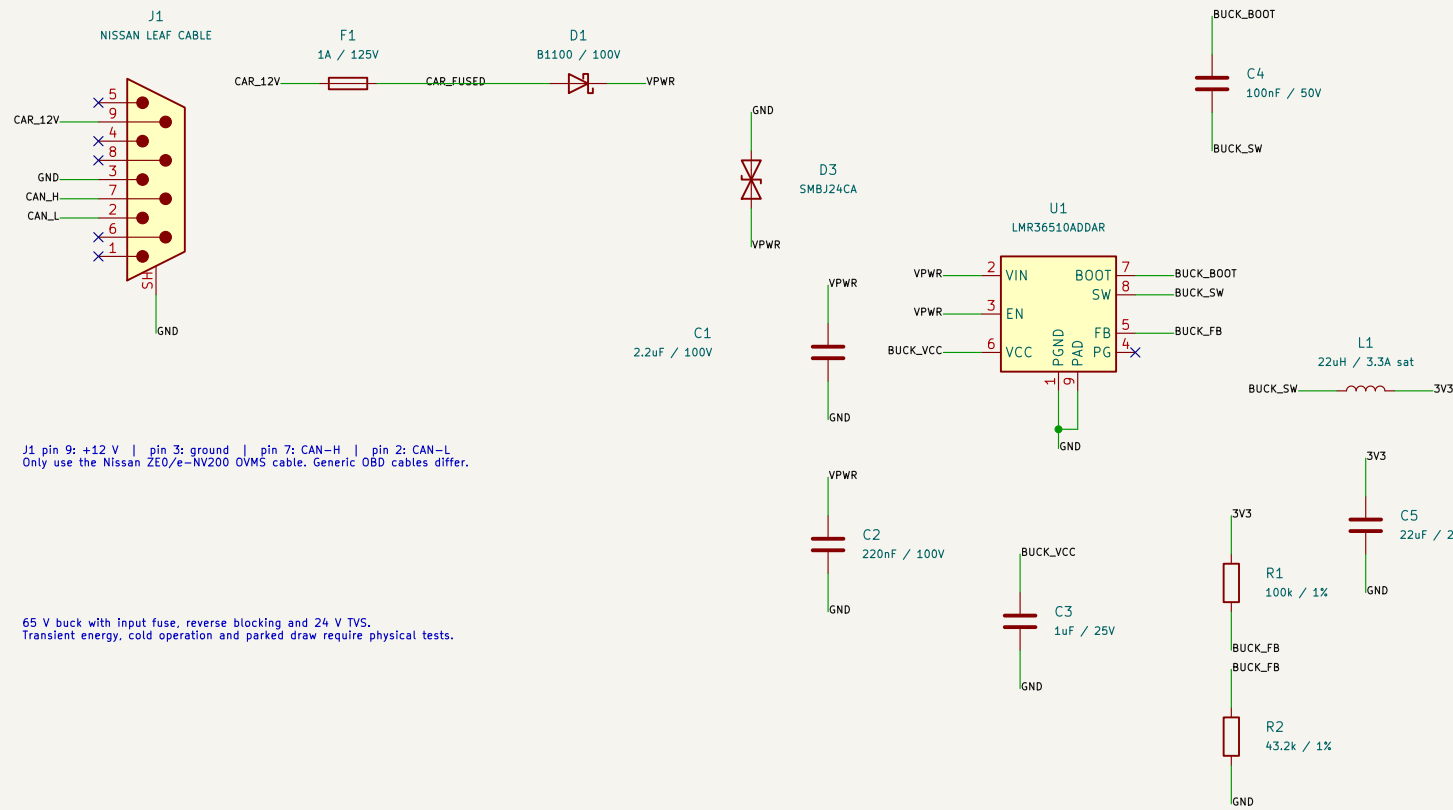


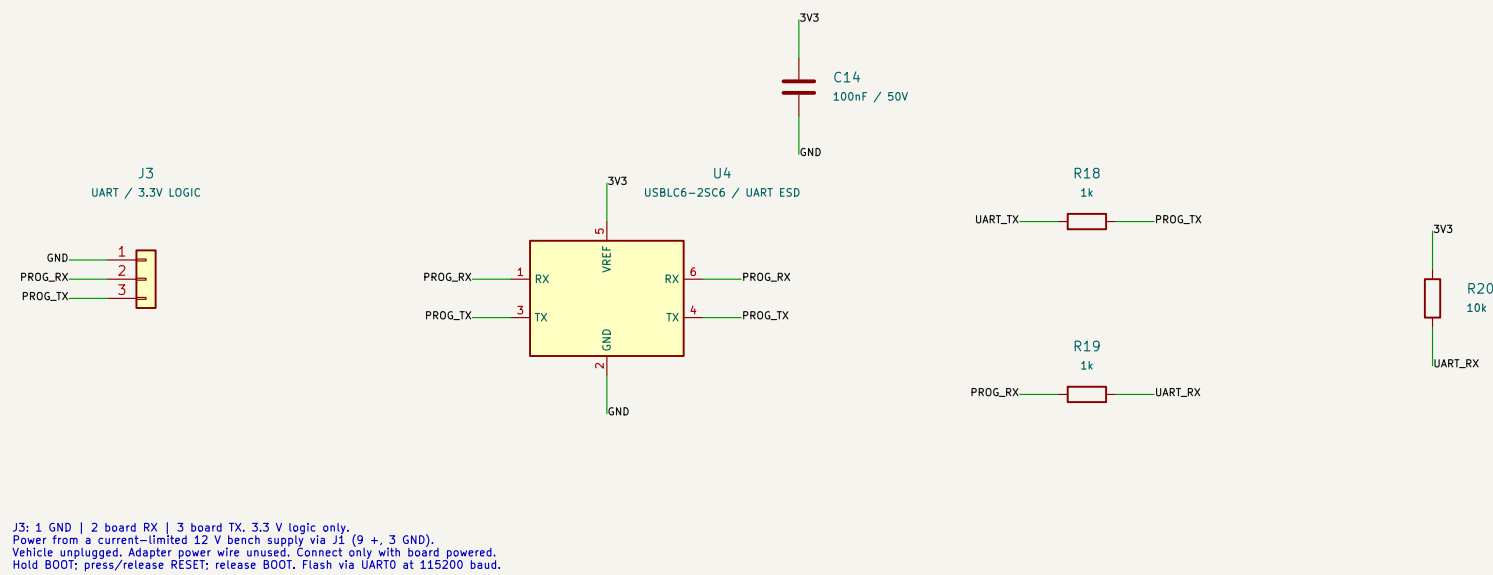
LEAF HEAT / Wi-Fi + one CAN channel

REV F – HAND-SOLDER PROTOTYPE – FACTORY: U1 / U2 / U4 / L1

1 VEHICLE POWER / 3.3 V SUPPLY



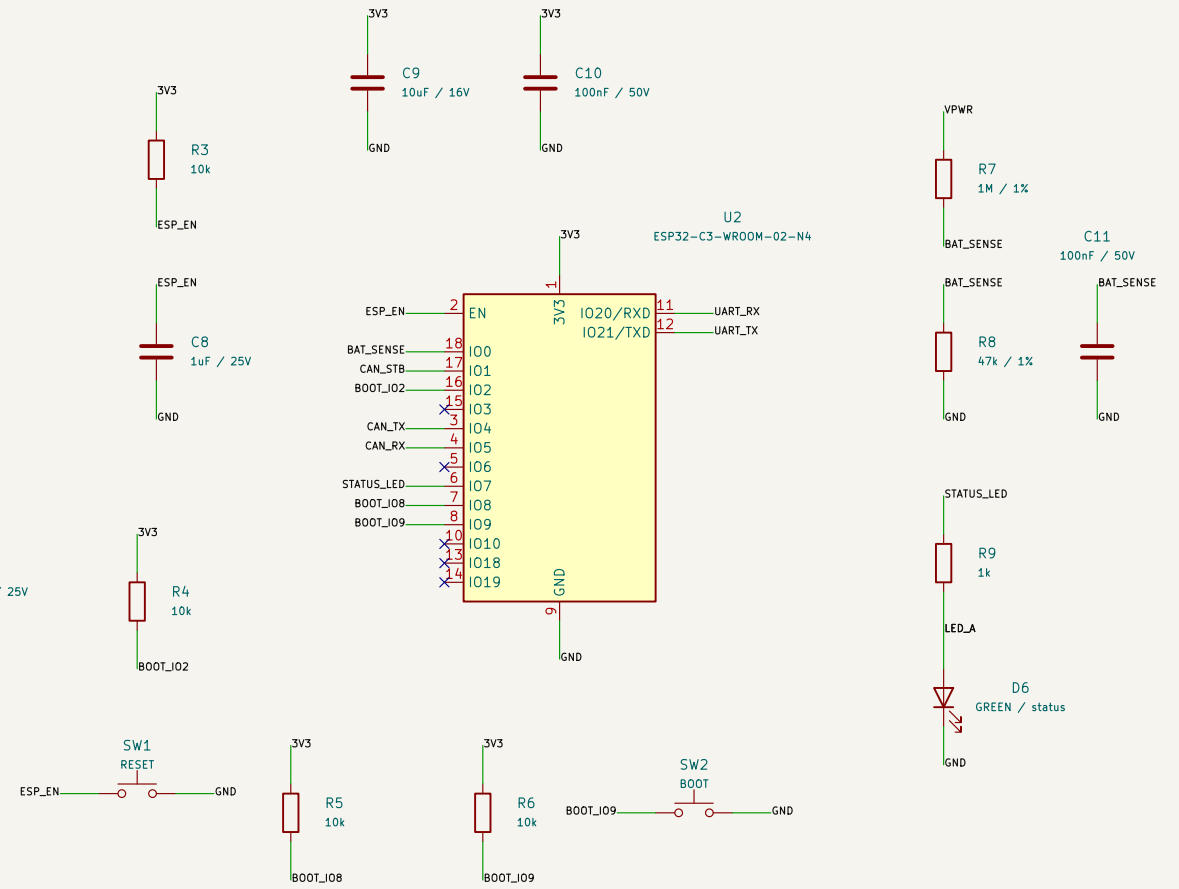
3 UART PROGRAMMING – NO USB / NO PROGRAMMER POWER



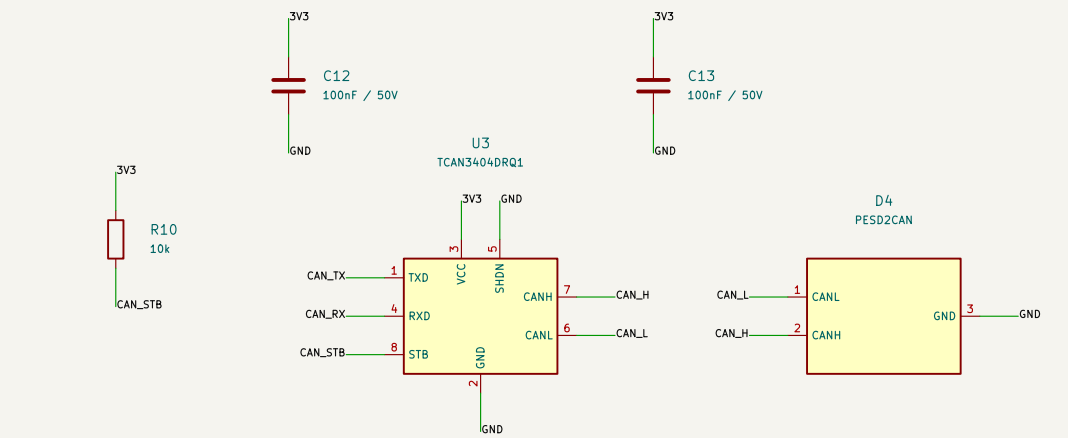
J3: 1 GND | 2 board RX | 3 board TX. 3.3 V logic only.
Power from a current-limited 12 V bench supply via J1 (9 +, 3 GND).
Vehicle unplugged. Adapter power wire unused. Connect only with board powered.
Hold BOOT; press/release RESET; release BOOT. Flash via UART0 at 115200 baud.

GPIO3 / GPIO18 / GPIO19 unused. No USB-presence sensing in Rev F.
OTA updates need firmware implementation. UART remains the recovery interface.

2 Wi-Fi / BOOT / BATTERY SENSE



4 VEHICLE CAN – NO TERMINATION



CAN TX: GPIO4 | RX: GPIO5 | STB: GPIO1
STB is pulled high during reset. Drive low only for a requested action.
500 kbit/s, standard 11-bit frames; 2013-2015 Leaf profile to be confirmed.
TCAN3404: SHDN pin 5 tied to GND. C13 remains an extra rail bypass.

Voltage sense is after the input diode; calibration required.
Software low-battery sleep is required; there is no hardware battery disconnect.

Engineering review and vehicle tests required before release

DIY Leaf project

Sheet: /
File: leaf-heat-v6.kicad_sch

Title: Leaf Heat – Wi-Fi climate control

Size: A3 Date: 2026-09-12

SIZE: A3	Date: 2020-09-12	REV: 1
KiCad E.D.A. 10.0.6		Id: 1/1

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